

Prior learning

SL & HL

Parallelogram's area	$A = bh$, b = base, h = height
Triangle's area	$A = \frac{1}{2}(bh)$, b = base, h = height
Trapezoid's area	$A = \frac{1}{2}(a + b)h$, a, b = parallel sides, h = height
Circle's area	$A = \pi r^2$, r = radius
Circle's circumference	$C = 2\pi r$, r = radius
Cuboid's volume	$V = lwh$, l = length, w = width h = height
Cylinder's volume	$V = \pi r^2 h$, r = radius, h = height
Prism's volume	$V = Ah$, A = cross-section area h = height
Cylinder curve's area	$A = 2\pi rh$, r = radius, h = height
Distance between 2 points $(x_1, y_1), (x_2, y_2)$	$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
Coordinates for midpoints	$(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$

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Solutions of a quadratic equation in the form $ax^2 + bx + c = 0$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad a \neq 0$
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Topic 1: Number and algebra

SL & HL

Arithmetic sequence nth term	$u_n = u_1 + (n - 1)d$
Arithmetic sequence sum of n terms	$S_n = \frac{n}{2}(2u_1 + (n - 1)d) = \frac{n}{2}(u_1 + u_n)$
Geometric sequence nth term	$u_n = u_1 \cdot r^{n-1}$
Geometric sequence sum of n terms	$S_n = u_1 \cdot \frac{r^n - 1}{r - 1} = u_1 \cdot \frac{1 - r^n}{1 - r}, \quad r \neq 1$
Compound interest	$FV = PV \times \left(1 + \frac{r}{100k}\right)^{kn}$ FV = future value, PV = present value, n = # of years, k = # of compounding periods/year, r% = nominal annual rate of interest
Exponents and logarithms equivalence	$a^x = b \iff x = \log_a b, \quad a, b > 0, a \neq 1$
Percentage error	$\varepsilon = \left \frac{v_A - v_E}{v_E} \right \times 100\%$ v_A = approximate value, v_E = exact value

Topic 1: Number and algebra

HL

Law of logarithms for $a, x, y > 0$	$\log_a(xy) = \log_a x + \log_a y,$ $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y,$ $\log_a(x^m) = m \log_a x.$
The sum of an infinite geometric sequence	$S_\infty = \frac{u_1}{1 - r}, \quad r < 1$
Complex numbers	$z = a + bi$
Discriminant	$\Delta = b^2 - 4ac$
Modulus-argument (polar) & Exponential (Euler) form	$z = r(\cos \theta + i \sin \theta) = re^{i\theta} = r \operatorname{cis} \theta$
Determinant of a 2x2 matrix	$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \det A = A = ad - bc$
Inverse of a 2x2 matrix	$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \det A \neq 0$
Power formula for a diagonalizable matrix	$M^n = PD^nP^{-1}$, where P is the matrix of eigenvectors and D is the diagonal matrix of eigenvalue

Topic 2: Functions

SL & HL

Equations of a straight line	$y = mx + c; \quad ax + by + d = 0$ $y - y_1 = m(x - x_1)$
Gradient formula	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Axis of symmetry of a quadratic function	$f(x) = ax^2 + bx + c \Rightarrow x = -\frac{b}{2a}$

Topic 2: Functions

HL

Logistic function	$f(x) = \frac{L}{1 + Ce^{-kx}}, \quad L, k, C > 0$
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Topic 3: Geometry & trigonometry		SL & HL
Distance between two points $(x_1, y_1, z_1), (x_2, y_2, z_2)$	$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$	
Coordinates of midpoint $(x_1, y_1, z_1), (x_2, y_2, z_2)$	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$	
Volume: Right-pyramid	$V = \frac{1}{3}Ah, \quad A = \text{base area}, h = \text{height}$	
Volume: Right cone	$V = \frac{1}{3}\pi r^2 h, \quad r = \text{radius}, h = \text{height}$	
Area: Cone curve	$A = \pi rl, \quad r = \text{radius}, l = \text{slant height}$	
Volume: Sphere	$V = \frac{4}{3}\pi r^3, \quad r = \text{radius}$	
Surface area: Sphere	$A = 4\pi r^2, \quad r = \text{radius}$	
Sine rule	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	
Cosine rule	$c^2 = a^2 + b^2 - 2ab \cos C$ $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$	
Area: Triangle	$A = \frac{1}{2}ab \sin C$	
Length of an arc	$l = r\theta, \quad r = \text{radius}, \theta = \text{angle in radians}$	
Area of a sector	$A = \frac{1}{2}r^2\theta, \quad r = \text{radius}, \theta = \text{angle in radians}$	

Topic 3: Geometry & trigonometry		HL
Length of an arc	$l = r\theta$ $r = \text{radius}, \theta = \text{angle in radians}$	
Area of a sector	$A = \frac{1}{2}r^2\theta$ $r = \text{radius}, \theta = \text{angle in radians}$	
Identities	$\cos^2 \theta + \sin^2 \theta = 1$ $\tan \theta = \frac{\sin \theta}{\cos \theta}$	
Transformation matrices	$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$ ·reflection in the line $y = (\tan \theta)x$ $\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$ ·horizontal stretch by scale factor of k $\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$ ·vertical stretch with scale factor of k $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$, centre $(0, 0)$ ·enlargement with scale factor of k $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, anticlockwise rotation of angle θ about the origin ($\theta > 0$) $\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, clockwise rotation of angle θ about the origin ($\theta > 0$)	

Topic 3: Geometry & trigonometry		HL
Magnitude of a vector	$ \mathbf{v} = \sqrt{v_1^2 + v_2^2 + v_3^2}$	
Vector equation of a line	$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$	
Parametric form of a line	$x = x_0 + \lambda l, \quad y = y_0 + \lambda m, \quad z = z_0 + \lambda n$	
Scalar product	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3$ $\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \mathbf{w} \cos \theta$ where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$	
Angle between two vectors	$\cos \theta = \frac{v_1 w_1 + v_2 w_2 + v_3 w_3}{ \mathbf{v} \mathbf{w} }$	
Vector product	$\mathbf{v} \times \mathbf{w} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$ $ \mathbf{v} \times \mathbf{w} = \mathbf{v} \mathbf{w} \sin \theta$ where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$	
Area of a parallelogram	$A = \mathbf{v} \times \mathbf{w} $, where $\mathbf{v}$ and $\mathbf{w}$ form two adjacent sides	

Topic 4: Statistics & probability		SL & HL
Interquartile range	$IQR = Q_3 - Q_1$	
Mean, $\bar{x}$, of a set of data	$\bar{x} = \frac{\sum_{i=1}^k f_i x_i}{n}$, where $n = \sum_{i=1}^k f_i$	
Probability of an event A	$P(A) = \frac{n(A)}{n(U)}$	
Complementary events	$P(A) + P(A') = 1$	
Combined events	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	
Mutually exclusive events	$P(A \cup B) = P(A) + P(B)$	
Conditional probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	
Independent events	$P(A \cap B) = P(A)P(B)$	
Expected value: Discrete random variable X	$E(X) = \sum xP(X = x)$	
Binomial distribution <i>Mean; Variance :</i>	$X \sim B(n, p)$ $E(X) = np; \text{Var}(X) = np(1 - p)$	

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Topic 4: Statistics & probability		HL
Linear transformation of a random variable	$E(aX + b) = aE(X) + b$ $Var(aX + b) = a^2Var(X)$	
Linear combinations of n independent random variables, $X_1, X_2, \dots, X_n$	$E(a_1X_1 \pm a_2X_2 \pm \dots \pm a_nX_n) = a_1E(X_1) \pm a_2E(X_2) \pm \dots \pm a_nE(X_n)$ $Var(a_1X_1 \pm a_2X_2 \pm \dots \pm a_nX_n) = a_1^2Var(X_1) + a_2^2Var(X_2) + \dots + a_n^2Var(X_n)$	
Unbiased estimate of population variance	$s_{n-1}^2 = \frac{n}{n-1}s_n^2$ (Sample statistics)	
Poisson distribution — Mean ; Variance	$X \sim \text{Po}(m)$ $E(X) = m$; $Var(X) = m$	
Transition matrices	$T^n \mathbf{s}_0 = \mathbf{s}_n$, where $\mathbf{s}_0$ is the initial state	

Topic 5: Calculus		SL & HL
Derivative of x^n	$f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$	
Integral of x^n	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$	
Area between curve $y = f(x)$ and x-axis	$A = \int_a^b y dx, \quad \text{where } f(x) > 0$	
The trapezoidal rule where $h = \frac{b-a}{n}$	$\int_a^b y dx \approx$ $\frac{1}{2}h(y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1}))$	

Topic 5: Calculus		HL
Derivative of $\sin x$	$f(x) = \sin x \Rightarrow f'(x) = \cos x$	
Derivative of $\cos x$	$f(x) = \cos x \Rightarrow f'(x) = -\sin x$	
Derivative of e^x	$f(x) = e^x \Rightarrow f'(x) = e^x$	
Derivative of $\ln x$	$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$	
Chain rule	$y = g(u), u = f(x) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$	
Product rule	$y = uv \Rightarrow \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$	
Quotient rule	$y = \frac{u}{v} \Rightarrow \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$	
Acceleration	$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$	
Distance; Displacement travelled from t_1 to t_2	$\text{dist} = \int_{t_1}^{t_2} v(t) dt; \quad \text{disp} = \int_{t_1}^{t_2} v(t) dt$	
Standard integrals	$\int \frac{1}{x} dx = \ln x + C$	

Standard integrals	$\int \sin x dx = -\cos x + C$
	$\int \cos x dx = \sin x + C$
	$\int e^x dx = e^x + C$
Area enclosed by a curve and x-axis	$A = \int_a^b y dx$
Area enclosed by a curve and y-axis	$A = \int_a^b x dy$
Volume of revolution about x or y axes	$V = \int_a^b \pi y^2 dx \quad \text{or} \quad V = \int_a^b \pi x^2 dy$
Acceleration	$a = \frac{dv}{dt} = \frac{d^2s}{dt^2} = v \frac{dv}{ds}$
Distance; displacement from t_1 to t_2	$\text{dist} = \int_{t_1}^{t_2} v(t) dt \quad ; \quad \text{disp} = \int_{t_1}^{t_2} v(t) dt$
Euler's method	$y_{n+1} = y_n + h \times f(x_n, y_n); \quad x_{n+1} = x_n + h$ where h is a constant
Euler's method for coupled systems	$x_{n+1} = x_n + h \cdot f_1(x_n, y_n, t_n)$ $y_{n+1} = y_n + h \cdot f_2(x_n, y_n, t_n)$ $t_{n+1} = t_n + h$ where h is a constant (step length)
Exact solution for coupled linear differential equations	$\mathbf{x} = Ae^{\lambda_1 t} \mathbf{p}_1 + Be^{\lambda_2 t} \mathbf{p}_2$

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